Esteban S. Zamora

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Education

Vanderbilt University Nashville, TN
B.E. Electrical & Computer Engineering | Minor: Engineering Management | GPA: 3.381 *Aug. 2023 – May 2027*

- Concentrations: Embedded Computing, Cyber-Physical Systems, Microelectronics
- Honors: Dean's List, Hispanic Scholarship Fund Scholar, Crecere Aude Scholar
- Coursework: Microelectronic Systems, Intermediate Software Design, Data Structures & Algorithms, Microcontrollers, Digital Systems, FPGA Design, DSP, Electronics I

Experience

Embedded Software Research Assistant Jan. 2026 – Present
BioMIID Lab – Vanderbilt University *Nashville, TN*

- Developed data acquisition software in C for a live-cell microscopy system, architecting a full pipeline from 4-sensor I2C polling on an Arduino Mega, through a 16-bit AD5669 DAC, into a calibrated NI PCIe-6353 DAQ, ensuring accurate real-time readings for researchers.
- Configured NIS-Elements as a real-time data acquisition GUI, setting up analog input channels, custom decoding equations, and live time-series plots to display calibrated sensor readings to researchers.
- Validated the full signal chain through structured integration tests and systematic fault isolation, resolving a firmware safety-timeout bug, NI-DAQ channel mismatches, scaling equation errors, and TTL pulse timing issues confirmed via oscilloscope.

Electrical Engineering Intern May 2025 – Aug. 2025
Novelis *Guthrie, KY*

- Programmed and validated safety-critical PLC software (RSLogix 5000) for motor control systems, implementing inspection interlocks and sequenced remote restart logic deployed on live industrial production floors.
- Developed HMIs in FactoryTalk View Studio for real-time monitoring of rolling lines, furnaces, and conveyor systems.

Digital Systems Teaching Assistant Aug. 2025 – Dec. 2025
ECE Department – Vanderbilt University *Nashville, TN*

- Debugged student SystemVerilog designs (ALUs, FSMs, datapaths) in Quartus and ModelSim using waveform analysis and expected-vs-observed verification methodology.
- Guided 30+ students through verification-driven lab workflows, diagnosing timing, reset, and state-machine faults across simulation and FPGA implementations.

Electrical & Design Engineering Technician Jan. 2024 – Present
The Wond'ry – Vanderbilt University *Nashville, TN*

- Built functional prototypes for VU Medical Center and Nissan Motor Corporation using 3D printers, laser cutters, and PCB fabrication tools.
- Advised 100+ students and faculty per week on embedded systems design, sensor integration, and software debugging.

Projects

Advanced Soil Sensor System | *C, Arduino, RS-485, Modbus-RTU, LCD, Git, Team Lead* Feb. 2025 – Present

- Led firmware development implementing Modbus-RTU over RS-485 to poll an NPK sensor and stream live calibrated readings to an LCD display; validated accuracy against certified soil references, achieving 88.9% measurement accuracy.
- Directed parallel fault isolation across the team, assigning signal timing, firmware data handling, and bus termination to separate members, tracing the root cause to incorrect register addresses in the voltage decoding logic.

Gravitational Simulation | *C++, OOP, Design Patterns, Git, CI/CD* Fall 2025

- Implemented a 14-body solar system simulation in C++ applying Singleton, Visitor, Factory, and Prototype design patterns; correctly reproduced Earth's full year-long orbit validated at every 100-second interval across 31.5 million time steps.
- Passed a 5-test automated suite (gravitational force, Newton inertia, factory validation, full solar system, and free-body motion) with position accuracy within 1000 km; enforced code quality via clang-format linting and structured Git commits.

Autonomous Maze-Solving Robot | *C, ATmega32U4, PID, FSM, Embedded Control* Nov. 2025

- Implemented a full embedded software stack in C: PID controller for smooth line-following, FSM for intersection classification, and reflectance sensor driver logic on an ATmega32U4.
- Verified correctness through iterative hardware-in-the-loop testing, isolating overshoot, oscillation, and misclassification bugs via systematic PID gain tuning and firmware fault analysis.

Speaker Recognition System | *MATLAB, Signal Processing, Data Analysis* Fall 2025

- Built a speaker identification system in MATLAB processing voice recordings to generate per-subject feature databases and classify unknown speakers by minimum-distance scoring.
- Validated identification accuracy across a full speaker test matrix, analyzed misclassification patterns, and documented results in a structured comparison spreadsheet.

Technical Skills

Programming: C, C++, Python, Java, Assembly, Verilog, VHDL, JavaScript, HTML, CSS
Software Engineering: OOP design patterns, unit & integration testing, CI/CD pipelines, Git, data acquisition GUI development, hardware-software interface development
Embedded Interfaces: RS-485 (Modbus-RTU), UART, I2C (TWI), SPI, PWM, GPIO, timer- and interrupt-driven peripherals
Tools: NI-DAQmx, NIS-Elements 6.20, Arduino IDE, Microchip Studio, Quartus Prime, ModelSim, RSLogix 5000, FactoryTalk Studio, KiCAD, LTspice, LabVIEW, COMSOL, SolidWorks, Fusion 360